7 A particular X-ray tube uses molybdenum (Mo) as the target element and another X-ray tube uses tungsten (W). An accelerating potential of 25 kV is applied to both tubes, giving rise to continuous spectrums being formed. The atomic number Z of molybdenum is 42 while that of tungsten is 74.

(a) Explain, with reference to the mechanism of X-ray production,

(i) how the continuous spectrum is formed, and

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.....

.....

..... [2]

(ii) why the minimum wavelength produced is the same for both target elements.

……

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……

…… [2]

(b) Characteristic peaks K_α and K_β occur for molybdenum, but not for tungsten at an accelerating potential of 25 kV. In order to obtain the characteristic spectra for tungsten, the accelerating potential has to be increased beyond 25 kV.

Explain

(i) why the intensity of the K_α X-ray is typically greater than the K_β X-ray for molybdenum.

……

…… [1]

(ii) why the characteristic spectra for tungsten only appear when the accelerating potential is greater than that necessary to produce characteristic spectra for molybdenum.

……

……

…… [2]

(c) The X-ray spectrum of molybdenum has a particular characteristic spectral line of wavelength 6.6×10^{-11} m, produced by electrons making transitions between two energy levels of the molybdenum atom.

Calculate, in electron-volts, the energy of an X-ray photon of wavelength 6.6×10^{-11} m.

energy = …… eV [2]